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Differential rotations of north Kyushu Island related to middle Miocene clockwise rotation of SW Japan

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Abstract. Paleomagnetic measurements were made on samples of Cretaceous granites and Tertiary sedimentary rocks from north Kyushu Island in order to clarify the kinematics of clockwise (CW) rotation of the SW Japan tectonic block at 15 Ma. A paleomagnetic direction from Oligocene sedimentary rocks in the northern part of north Kyushu Island indicates a significant CW rotation of the region (R=35.9°± 11.0°) relative to eastern Asia, while the direction from Oligocene to early Miocene sedimentary rocks in the western part of north Kyushu Island shows no significant tectonic displacement. In situ directions of secondary magnetization from Eocene sedimentary rocks in the southwestern part of north Kyushu Island, and stable remanences from Cretaceous granites, provide CW deflections of declination. These paleomagnetic data, combined with those previously reported, suggest that the main part of north Kyushu Island (Main Kyushu) was rotated CW (36°) between 30 and 14 Ma relative to eastern Asia, whereas no significant tectonic displacement has taken place since 20 Ma in the western part (Western Kyushu). Considering regional geological structures, the paleomagnetic results indicate that the western extent of the SW Japan tectonic block which experienced middle Miocene CW rotation is truncated west of Main Kyushu. The paleomagnetic data also suggest that Main Kyushu rotated counterclockwise (28° ± 13°) relative to the main part of SW Japan between 30 and 14 Ma. This result implies that the westernmost part of the SW Japan block (Main Kyushu) did not experience as much rotation as its main part during CW rotation of the whole block.

Introduction

The Japanese Islands consist mainly of late Paleozoic to Cenozoic accretionary complexes, and their metamorphosed equivalents, which formed in a convergent plate boundary along the eastern margin of the Asian continent [e.g., *Isozaki and Maruyama*, 1991]. During formation of the Japan Sea back-arc basin in late Oligocene to middle Miocene time [*Tamaki et al.*, 1992; *Ingle*, 1992], the Japanese Islands were rifted and then transported away from the Asian continent as continental slivers. Many kinematic reconstructions have been proposed for the opening of the Japan Sea and the associated displacement of the Japanese Islands [e.g., *Otofuji et al.*, 1985a; *Jolivet et al.*, 1991]. Most reconstructions suggest that Southwest (SW) Japan and Northeast (NE) Japan (Figure 1a) were subjected to differential rotations in Miocene time [*Hayashida et al.*, 1991; *Jolivet et al.*, 1991; *Jolivet and Tamaki*, 1992; *Otofuji and Matsuda*, 1987; *Otofuji et al.*, 1985a, b, 1991, 1994; *Tosha and Hamano*, 1988]. Paleomagnetic results have documented middle Miocene clockwise (CW) rotation of SW Japan by approximately 50° at about 15 Ma [*Hayashida*, 1986; *Hayashida et al.*, 1991; *Otofuji et al.*, 1985b, 1991; *Otofuji and Matsuda*, 1987]. It has also been proposed that SW Japan acted as a coherent block during this CW rotation [*Otofuji et al.*, 1985b; *Otofuji and Matsuda*, 1987; *Otofuji et al.*, 1991].

The coherent block rotation model is based partly on the geometry of pre-Miocene geotectonic units of SW Japan, which show a zonal arrangement parallel to the present trench, the Nankai trough, with an approximately ENE- WSW trend (Figure 1a). No large fault zone cuts across the arc. Paleomagnetic data from the San'in to Tokai districts in the main part of SW Japan (Figure 1a) show consistent values of rotation, a result which argees with the coherent block rotation model. Paleomagnetic directions from the easternmost part of SW Japan, however, indicate smaller CW rotations than that of the main part [*Itoh*, 1988; *Itoh and Ito*, 1989]. This differential rotation is attributed to counterclockwise (CCW) rotation of the easternmost part relative to the main part of SW Japan caused by collision of the Izu- Bonin arc with the central part of the Japanese Islands in middle Miocene time [ca. 15-12 Ma; *Itoh*, 1988]. Refinements of the coherent block model have also been proposed based on structural investigations of SW Japan; *Kanaori* [1990] divides the inner zone of SW Japan, north of the Median Tectonic Line (Figure 1a), into approximately 10 blocks bounded by active faults and suggests that these small blocks might have been rotated during the formation of the Japan Sea. *Fournier et al.* [1995] and *Jolivet et al.* [1995] suggest that the documented CW rotation of SW Japan may reflect rotations of the whole of the SW Japan block as well as smaller blocks within this domain.

The aim of this paper is to use new paleomagnetic data in order to assess the coherent block model for the northern half of the Kyushu Island (north Kyushu Island). Pre- Miocene structures of SW Japan with ENE-WSW trends can be traced across north Kyushu Island, with the trend changing to a N-S direction in the southern part of the Kyushu Island (Figure 1a). The whole of north Kyushu Island has thus been interpreted as a part of the CW-rotated SW Japan block [e.g., *Otofuji et al.*, 1991]. Geological features and paleomagnetic data from the western part of north Kyushu Island imply the existence of discrete tectonic blocks with tectonic histories different from SW

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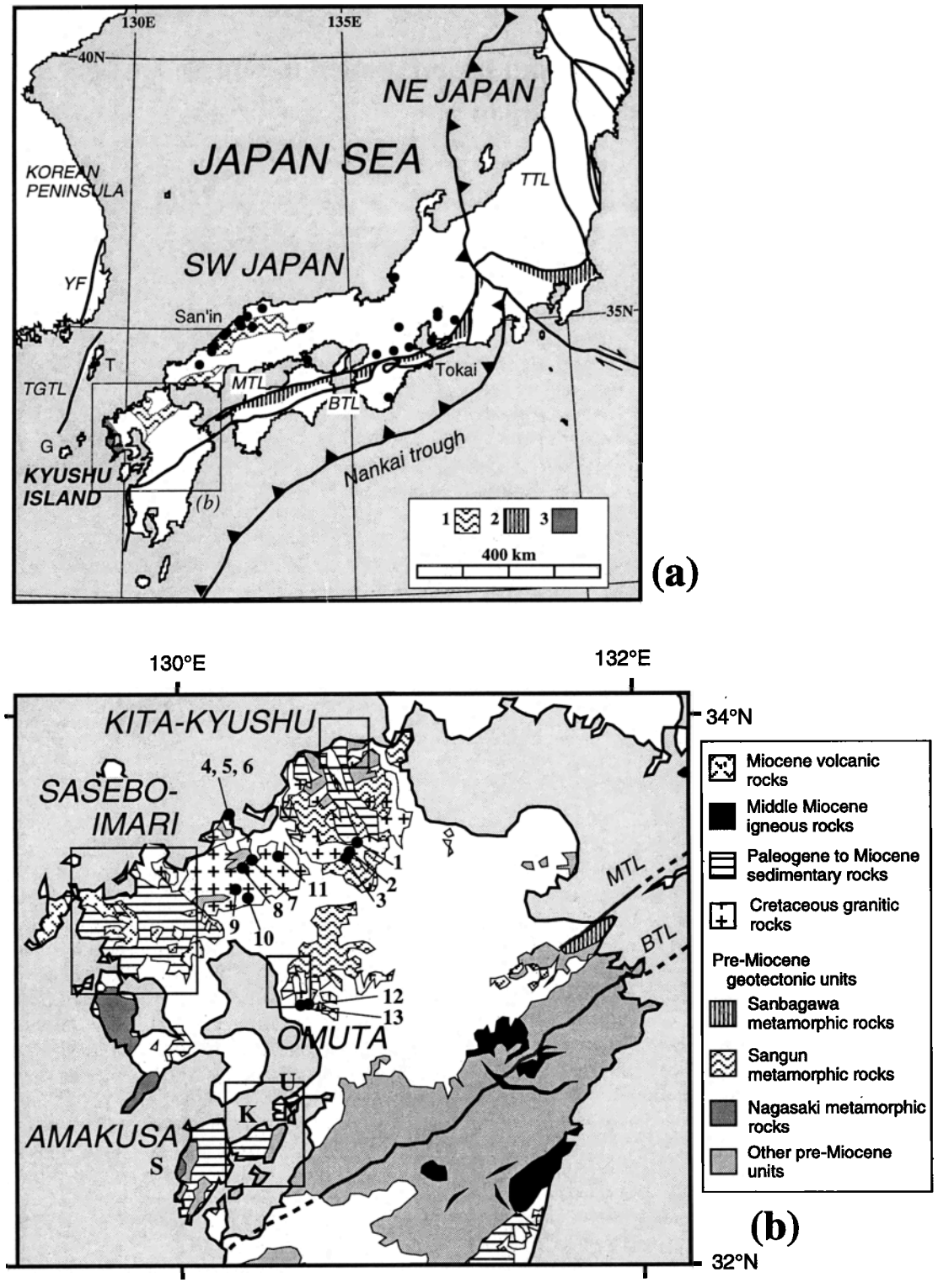


Figure 1. (a) Simplified geotectonic map of the Japanese Islands [after *Isozaki and Maruyama*, 1991]. Thick lines indicate major tectonic lines and plate boundaries; MTL, Median tectonic line; BTL, Butsuzo tectonic line; TTL, Tanakura tectonic line; TGTL, Tsushima-Goto tectonic line; YF, Yangsan Fault. MTL and BTL represent the general trend of the zonal structure of pre-Miocene geotectonic units in SW Japan. Solid circles denote localities of paleomagnetic sites which have shown a clockwise rotation of the coherent SW Japan block at 15 Ma [*Otofuji et al.*, 1985b; *Hayashida*, 1986; *Nakajima and Hirooka*, 1986; *Otofuji and Matsuda*, 1987; *Fukuma and Torii*, 1990; *Hayashida et al.*, 1991; *Otofuji et al.*, 1991]. 1, Sangun metamorphic rocks; 2, Sanbagawa metamorphic rocks; 3, Nagasaki metamorphic rocks; T, Tsushima Islands; G, Goto Islands. (b) Simplified geological map of north Kyushu Island [after *Geological Survey of Japan*, 1987]. Squares indicate paleomagnetic study areas. Solid circles with numerals denote sampling sites of Cretaceous granitic rocks. S, Shimoshima Island; K, Kamishima Island; U, Uto Peninsula.

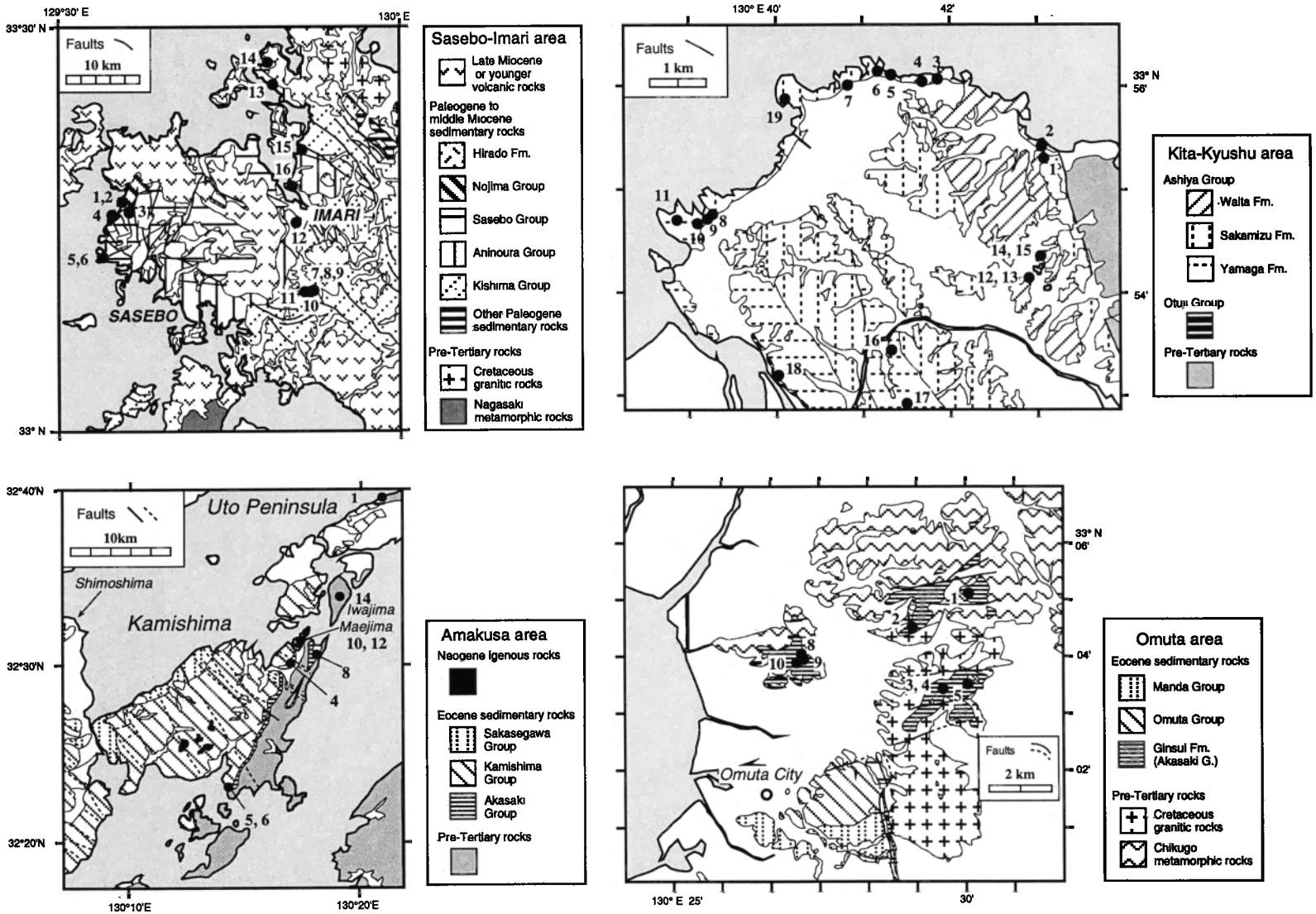


Figure 2. Locations of paleomagnetic sampling sites (solid circles with numerals). Geologic maps in the Sasebo-Imari, Kita- Kyushu, Amakusa, and Omuta areas are simplified from *Geological Survey of Japan* [1959, 1989], *Tsuchi et al.* [1987], *Miki* [1981], and *Fukuoka Prefecture* [1987], respectively. Chikugo metamorphic rocks in the Omuta area are equivalent to the Sangun metamorphic rocks.

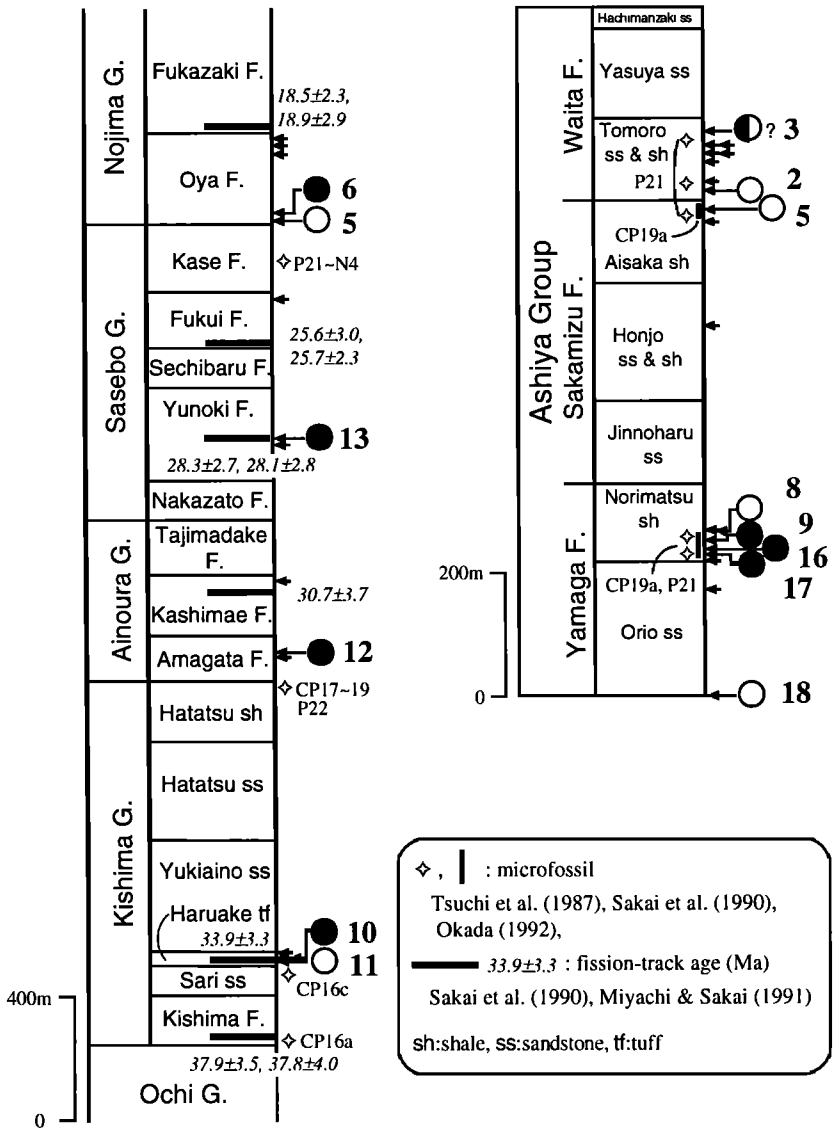


Figure 3. Stratigraphy of the (left) Sasebo-Imari area after *Imai et al.* [1958] and *Matsui et al.* [1989] and (right) Kita-Kyushu area after *Tsuchi et al.* [1987]. Horizons of paleomagnetic, biostratigraphic, and radiometric sampling are also shown. Arrows show horizons from which paleomagnetic samples were collected. Solid (open) circles indicate normal (reversed) paleomagnetic polarities. Site 3 in the Kita-Kyushu area displays a paleomagnetic direction inconsistent with the stable geomagnetic field direction at that locality (see text).

Japan. The pre-Miocene Nagasaki metamorphic rocks in the westernmost part of north Kyushu Island (Figure 1) show different structural trend from those of SW Japan [e.g., *Faure et al.*, 1988]. Paleomagnetic results from the Tsushima and Goto Islands in the Tsushima Strait area off the west coast of north Kyushu Island suggest CCW rotations of the two islands in middle Miocene time [*Ishikawa et al.*, 1989; *Ishikawa and Tagami*, 1991]. In this study, paleomagnetic measurements were carried out on samples from Cretaceous granitic rocks and Tertiary sedimentary rocks of north Kyushu Island (Figure 1) with the aim of clarifying tectonic motions of north Kyushu Island and their relationships with the CW rotation of SW Japan.

Geological Setting and Sampling

Pre-Miocene geotectonic units, which can be traced from the main part of SW Japan, are exposed on the northern and southern part of north Kyushu Island (Figure 1b) although the central part is covered by late Neogene to Quaternary volcanic rocks. Late Cretaceous granitic intrusions also crop out in the northern part. The Nagasaki metamorphic rocks are exposed on the western part of north Kyushu Island (Figure 1b) and show N-S to NE-SW structural trends [e.g., *Faure et al.*, 1988]. Paleogene sedimentary rocks overlie the pre-Tertiary rocks mentioned above (Figure 1b), while Miocene sedimentary rocks uncomfortably overlie the Paleogene sequences on the western part of north Kyushu Island. Paleomagnetic samples were collected from late Cretaceous granitic rocks and from the Paleogene and Miocene sedimentary rocks at four areas: the Sasebo-Imari, Kita-Kyushu, Amakusa, and Omuta areas (Figures 1b and 2).

In the Sasebo-Imari area, Paleogene to Miocene sedimentary sequences overlie the Nagasaki metamorphic rocks on the western side and are in fault contact with late Cretaceous granitic rocks on the eastern side (Figure 2). The sedimentary sequences are overlain by late Miocene Kita- Matsuura basalts. Non

deflected paleomagnetic directions were obtained from the basalts [*Creer and Ispir*, 1970; *Nomura*, 1967; *Ozima et al.*, 1968]. Paleomagnetic samples were collected at 16 sites from tuffs and mudstones of the Kishima Group, the Ainoura Group, the Sasebo Group, and the Oya Formation of the Nojima Group (Figures 2 and 3). Paleontological and geochronological data [*Miyachi and Sakai*, 1991; *Kano et al.*, 1991; *Okada*, 1992] show the ages of these units range from Oligocene to Early Miocene; from approximately 34 to 19 Ma (Figure 3).

Paleogene sequences on the northern part of north Kyushu Island can be divided into three groups, namely, the Nogata, Otsuji, and Ashiya Groups in ascending order. Paleomagnetic samples were collected at 19 sites from mudstones, fine sandstones, and tuffs of the Ashiya Group in the Kita-Kyushu area (Figures 2 and 3). *Tsuchi et al.* [1987] reported planktonic foraminifera assemblages assigned to Blow's Zone P.21 [*Blow*, 1969] from the Ashiya Group, and *Okada* [1992] correlated calcareous nannoplankton assemblages to Subzone CP19a of the nannofossil zonation (Figure 3). These micropaleontological data indicate that the entire Ashiya Group is of Oligocene age (ca. 30 Ma [*Tsuchi et al.*, 1987]).

Paleogene sequences on the southwestern part of north Kyushu Island (the Amakusa Islands and the Uto Peninsula) can be divided into three groups: the Akasaki, Kamishima, and Sakasegawa Groups in ascending order [Figure 2; *Miki*, 1981]. Microfossil data indicated that the Kamishima and Sakasegawa Groups are Middle to Late Eocene in age [*Yasuda*, 1984; *Okada*, 1992]. *Tashiro et al.* [1980] identified calcareous nannofossils assigned to early Middle Eocene from the Akasaki Group. Paleomagnetic samples of the Akasaki Groups were collected in the region from the Kamishima Island to Uto Peninsula, referred to as the Amakusa area in this paper. The Akasaki Group is composed mainly of purple-red mudstone and sandstone, and was sampled at eight sites (Figure 2). *Miki and Matsueda* [1985] shows that hematite in pigment form is responsible for the characteristic color and fills the interstices between grains. Eocene sequences on the Amakusa Islands and the Uto Peninsula are strongly folded with NNE-SSW trending fold axes. The fold axes are subhorizontal [*Miki*, 1975]. Various types of igneous rocks intrude into the folded strata. K-Ar ages were reported from these igneous rocks on the Shimoshima Island: 19 Ma from a granodiorite [*Shibata and Togashi*, 1975] and 14 to 7 Ma from andesite and basalt dikes [*Nagao et al.*, 1992]. The above age estimations of the sedimentary and igneous rocks constrain the phase of folding to have occurred sometime between late Eocene and early Miocene.

Paleogene sedimentary rocks in the Omuta area can be divided into three units [*Miki*, 1981]: the Ginsui Formation and the Omuta and Manda Groups in ascending order (Figure 2). *Okada* [1992] reported nannofossil assemblages assigned to CP14a (Middle Eocene) through to CP15b (Late Eocene) from the upper part of the Manda Group. The Ginsui Formation is composed mainly of purple-red mudstone and sandstone and is correlated to the Middle Eocene Akasaki Group in the Amakusa area [*Miki*, 1981]. Paleomagnetic samples from mudstones and sandstones of the Ginsui Formation were collected at eight sites (Figure2).

Paleomagnetic samples were collected from 13 sites in late Cretaceous granitic rocks (Figure 1b). The K-Ar ages of the granitic rocks range from 97 to 76 Ma with the peak occurrence at 90 Ma [e.g., *Nozawa*, 1970]. *Ito and Tokieda* [1986] obtained CW-deflected directions of magnetization from samples collected near sites 4, 5, and 6 of this paper.

Sedimentary rock samples were collected mainly using a gasoline-powered core drill or by hand sampling. Granitic rocks were collected by hand sampling. Six to 14 core or block samples oriented with a magnetic compass were collected from each site. One or two cylindrical specimens of 24 mm diameter and 22 mm height were prepared from each sample. Cubic specimens of about 20 mm were prepared from hand samples of sedimentary rock unsuitable for laboratory coring.

Paleomagnetic Analysis

Experiments

The stability of natural remanent magnetization (NRM) was examined by progressive thermal and alternating field (AF) demagnetization experiments. The NRM of each specimen was measured using a cryogenic magnetometer (ScT C-112) with a noise level of 2×10^{-11} Am^2. Thermal demagnetization was performed in air using a noninductively wound electric furnace in a four-layer mu- metal magnetic shield; the internal stray field was less than 5 nT. AF demagnetization was performed using a three-axes tumbler system contained in a three-layer mu-metal shield. Low-field magnetic susceptibility of pilot specimens from sedimentary rocks was also measured using a Bartington MS2 magnetic susceptibility meter at each thermal demagnetization step to monitor thermal alteration of samples.

Two or three pilot specimens from each site were firstly subjected to progressive demagnetization experiments. When the pilot demagnetization results of a site showed line trends of vector endpoints decaying toward the origin on vector endpoint diagrams [*Zijderveld*, 1967] or remagnetization circles on equal-area plots, remaining specimens of the site were demagnetized progressively using the peak fields or temperature values where pilot specimens' vector endpoints defined a linear trend or a remagnetization circle on the respective plot. AF demagnetization was used on the remaining specimens if pilot AF and thermal demagnetization isolated the same component of magnetization. The characteristic remanent magnetization (ChRM) of a specimen was determined by fitting a least squares line [*Kirschvink*, 1980] to vector endpoints displaying a linear trend toward the origin. The best fit line was anchored on the origin of the vector endpoint diagram. Site mean directions and associated statistical parameters were calculated after *Fisher* [1953] when the specimen ChRMs were defined by a linear decay of vector endpoints. The common direction shared by remagnetization circles was calculated using the method of *McFadden and McElhinny* [1988], where possible remagnetization circles were combined with directions determined from the linear decay of vector endpoints. The common direction was taken as a site mean of ChRMs. Site mean directions with α_{95} less than 20° were deemed reliable recorders of the ancient geomagnetic field and used in calculating group means.

Demagnetization Results

Sasebo-Imari area. Initial NRM intensities of specimens from the Sasebo-Imari area generally ranged from 10^{-9} to 10^{-8} Am^2, whereas specimens from sites 9 and 11 had exceptionally weak intensities of the order of 10^{-10} Am^2. Low-stability components with generally normal polarity, probably viscous overprints of the recent geomagnetic field direction, were removed in the demagnetization range below about 240°C and/or 10 mT. The low-stability components accounted for 50 to 90%

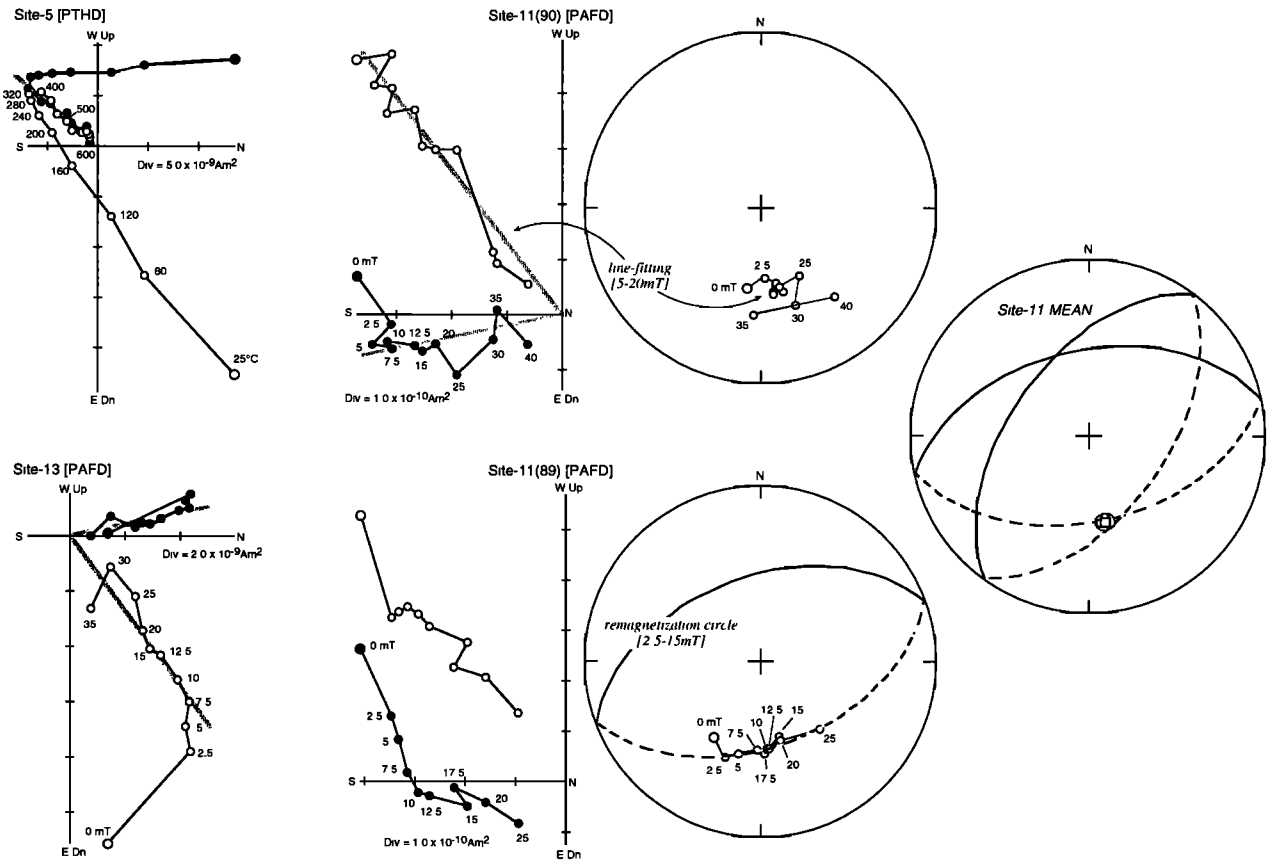


Figure 4a. Progressive demagnetization results from the Sasebo-Imari area shown on vector endpoint diagrams and equal-area plots. PTHD, progressive thermal demagnetization; PAFD, progressive alternating field demagnetization. On the vector endpoint diagram, solid and open circles are projections onto the horizontal and N-S vertical planes, respectively. On the equal-area plots, open symbols are on the upper hemispheres. The results from site 11 provided linear decays of the vector endpoints to the origin of the diagram and remagnetization circles. The site mean direction for site 11 (open square with oval of 95% confidence limit) was calculated by the method of *McFadden and McElhinny* [1988].

of the specimens' initial NRM intensity. For demagnetization levels above 240°C and/or 10 mT, five of 16 sites whose specimens had relatively strong initial NRM intensity (around 10^{-8} Am^2) yielded ChRMs. ChRMs were isolated in specimens from sites 5 and 13 by thermal demagnetization in the range between 360°C and 600°C (Figure 4a). Specimens from sites 6 and 12 provided ChRMs in the range from 240°C to 360°C or 470°C, while those from site 13 were in the range between 10 and 20 mT, with erratic magnetic behavior at higher demagnetization levels (Figure 4a). NRM intensities of specimens with a relatively weak initial value around 10^{-10} to 10^{-9} Am^2 were just above the noise level of the magnetometer after removal of the low-stability components, and these specimens showed unstable behavior of magnetization. Although specimens from site 11 had weak initial NRM intensities (~10^{-10} Am^2), four out of nine specimens of site 11, with relatively strong NRMs, showed stable magnetic behavior; two specimens showed roughly linear decay of vector end points between 10 and 25 mT, and two others provided remagnetization circles in the range from 5 to 15 mT (Figure 4a). These four specimens also showed unstable magnetic behavior above these demagnetization levels because of very weak intensities (<1 x 10^{-11} Am^2).

Site mean directions of ChRMs were finally determined from six sites (Table 1), which show an antipodal relationship with a north-to-south trend after tilt correction (Figure 5). Statistical parameters of the overall mean directions indicate that the site mean directions show a slightly better grouping after tilt correction (Table 1). Results of the fold test proposed by *McFadden* [1990] suggest that there is no reason to reject the hypothesis that the magnetization was acquired before the tilting occurred; for the in situ position, $\xi 2$=3.770, which exceeds the 95% confidence limit of 2.862, whereas for the tilt-corrected position, $\xi 2$=0.437, which does not exceed the 95% confidence limit (Table 1). The tilt-corrected directions probably represent the directions of the ancient geomagnetic field at the time of primary magnetization acquisition. The overall mean of the tilt-corrected directions (D=353.0°, I=52.9°, α_{95}=9.7°, and k=48.8) is regarded as a mean paleomagnetic direction of the Oligocene to early Miocene time in the Sasebo-Imari area.

Kita-Kyushu area. NRMs of specimens from the Kita-Kyushu area generally showed unstable behavior at demagnetization levels above 300°C and/or 15 mT; erratic directional and intensity changes were particularly clear above approximately 360°C and/or 25 mT. This instability of magnetizations meant that few of the specimens provided meaningful demagnetization results. However, demagnetization results from eight of 19 sites did indicate the existence of ChRMs isolated at demagnetization levels between about 200°C and

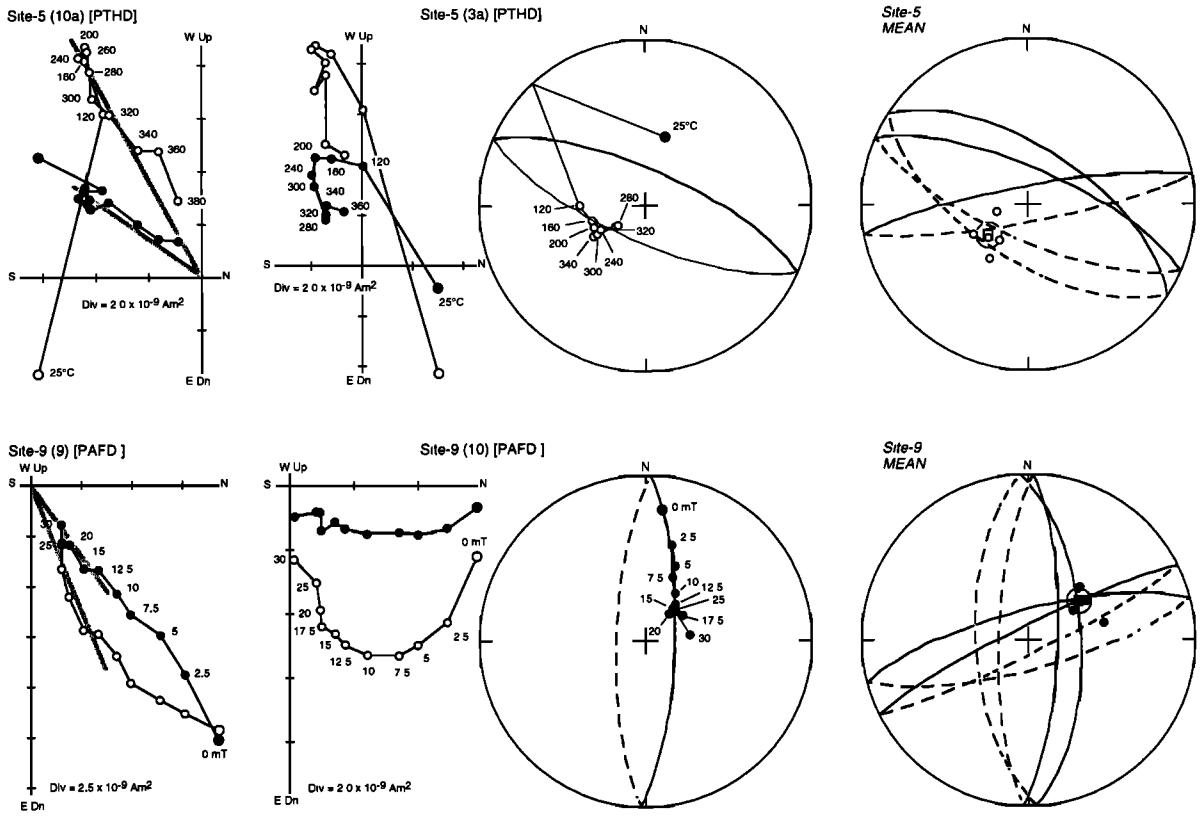


Figure 4b. Progressive demagnetization results from the Kita-Kyushu area shown on vector endpoint diagrams and equal-area plots. Solid symbols on the equal-area plots are on the lower hemispheres. Each site mean directions (square) was calculated using the method of *McFadden and McElhinny* [1988].

360°C and/or between about 10 and 25 mT (Figure 4b). The ChRMs of some specimens from the eight sites were determined from linear trends of vector endpoints decaying to the origin of the diagram. The other specimens yielded remagnetization circles, either because demagnetization spectra of the ChRM and the component with lower unblocking temperature or coercivity overlapped considerably, or because complete isolation of the ChRM was masked by unstable magnetic behavior above about 300°C and/or 15 mT (Figure 4b). A site mean direction of the ChRMs was calculated for each site based on the method of *McFadden and McElhinny* [1988] as shown in Figure 4b. Unstable magnetic behaviors during thermal demagnetization is probably due to the creation of a new magnetic phase. Initial susceptibility of specimens, except for tuff samples, increased remarkably at about 280°C or 360°C during thermal demagnetization. Viscous remanences (VRMs) were usually acquired in the laboratory after heating above 300°C, and were detected by a decay of the total magnetization within the magnetometer. The new magnetic phase may be magnetite formed by thermal alteration of clay minerals. The presence of clay minerals was detected by X ray diffraction analysis on minerals separated by a small magnet, although the presence of ferromagnetic minerals was not detected. In addition, the strong-field thermomagnetic results from the separated minerals (0.8 T in argon gas) indicated formation of magnetite on heating. The heating curve showed paramagnetic decay of strong-field magnetization below about 350°C and a subsequent increase of the magnetization with a peak at about 450°C, followed by a decrease of the magnetization up to 670°C. The cooling curve showed the Curie temperature of magnetite and a considerable increase of the magnetization. Unstable magnetic behavior was also observed during AF demagnetization and its cause is unknown. Median destructive fields (MDF) of NRMs for the Ashiya Group specimens were generally 5 to 10 mT, implying large grains of magnetic minerals are the principal carriers of NRM. Such minerals might have acquired anhysteresis remanences during AF treatments.

Site mean directions from the eight sites are shown in Figure 5 and listed in Table 1. The anomalous direction of site 3 may be attributed to a record of the transitional field during a geomagnetic reversal or to a local tectonic disturbance. The latter is considered less likely because the bedding attitude at site 3 is identical with the general trend of the Ashiya Group. All site means except that of site 3 show an antipodal relationship with the NE-SW trend after tilt correction. Tilt-corrected directions of the seven sites show a better grouping than in situ directions. Results of McFadden's [1990] fold test suggest that the magnetizations would have been acquired before tilting; $\xi 2$ in the in situ position (4.959) exceeded the 99% confidence limit of 4.253, while $\xi 2$ in the tilt-corrected position (1.496) did not exceed the 95% confidence limit of 3.086 (Table 1). A mean of the seven tilt-corrected directions (D=35.9°, I=45.3°, α_{95}=9.9°, and k=37.8) is thus regarded as the paleomagnetic direction of the Oligocene Ashiya Group at around 30 Ma for the Kita-Kyushu area.

Amakusa area. ChRMs were isolated by progressive thermal demagnetization for specimens from five of eight sites. A large fraction of the ChRM was unblocked at temperatures above

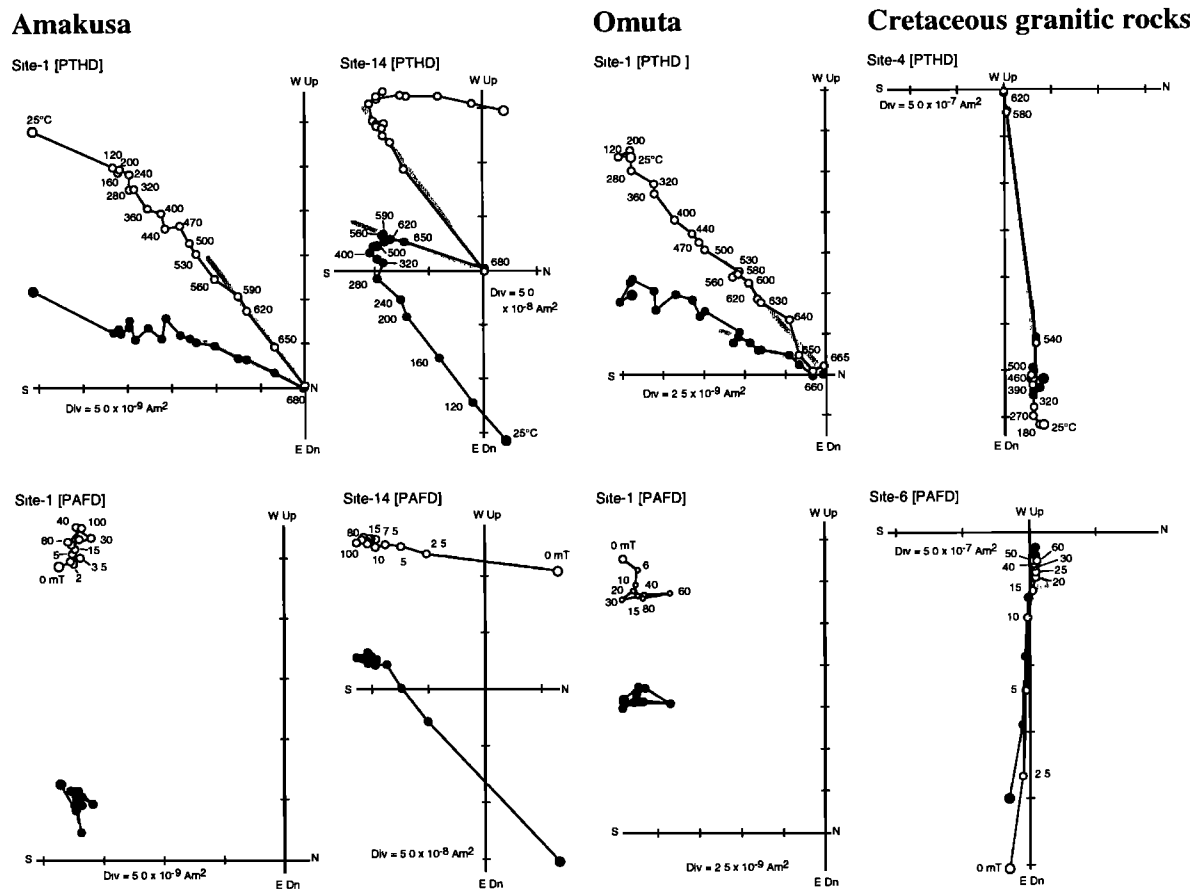


Figure 4c. Progressive demagnetization results from the (left) Amakusa and (middle) Omuta areas and (right) Cretaceous granitic rocks shown on vector endpoint diagrams.

580°C, while AF demagnetization in peak fields up to 100 mT was not sufficient to demagnetize the ChRMs (Figure 4c). These demagnetization results suggest that hematite is the principal magnetic carrier. Pilot specimens from the other three sites showed unstable behavior of NRM in the demagnetization levels above about 400°C and/or 20 mT, although some of the specimens showed linear trends of vector endpoints which did not decay toward the origin of the diagrams below these demagnetization levels. Strong-field thermomagnetic curves up to 700°C (0.85 T in argon gas) of bulk samples from the Akasaki Group showed irreversible behavior in the temperature range below 600°C; the strong-field magnetization on cooling was smaller than that on heating. The heating curve showed a broad decrease of the magnetization below about 550°C, except for a sharp decrease around 650°C observed in both the heating and cooling runs. This result possibly indicates transformation of maghemite to hematite during heating. The unstable magnetic behavior above 400°C may be due to thermal alteration of maghemite, although any difference was not found in the thermomagnetic curves between specimens with ChRMs and those with unstable NRMs. Site mean directions of ChRMs from the five sites show a tight cluster with a SW declination and negative inclination before tilt correction, while the tilt-corrected directions are more scattered (Figure 5). The mean in situ direction is D=210.0°, I=-50.7°, α_{95}=6.4°, and k=145.6 (Table 1) and is considered a secondary component acquired after regional folding in the Amakusa area. Judging from the secondary origin of hematite in the Akasaki Group [*Miki and Matsueda*, 1985], the component carried by hematite is probably a secondary chemical remanence.

Omuta area. Initial NRM intensities of specimens from site 1 were about 2×10^{-8} Am^2, while specimens from the other sites showed variable intensities ranging between 10^{-10} and 10^{-8} Am^2 for each site. Only specimens from site 1 were found to have ChRMs carried by hematite; these ChRMs were isolated between 500°C and 640°C, while NRM directions and intensities of the specimens did not change during progressive AF demagnetization (Figure 4c). On the other hand, pilot specimens from the remaining sites showed erratic directional and intensity changes above about 280°C to 400°C and/or above 10 to 20 mT. During thermal demagnetizations, an abrupt increase of initial susceptibility at about 280°C to 400°C and remarkable VRM acquisition in the laboratory were observed for these specimens. Magnetic behavior in strong-field thermomagnetic experiments (0.85 T in argon gas) of bulk samples were similar to that of the Akasaki Group, possibly indicating the existence of maghemite. There was no difference in thermomagnetic results between the stable and unstable sites. Inversion of maghemite is one of the ways in which hematite is formed in red sedimentary rocks [*Larson and Walker*, 1975]. The stability of NRMs of purple-red sedimentary rocks in the Omuta area, as well as the Akasaki Group in the Amakusa area, may depend on the amount of maghemite and/or the degree of the maturity of the inversion process. The in situ direction of site 1 has SW declination with

Table 1. Paleomagnetic Results From the Sasebo-Imari, Kita-Kyushu, Amakusa, and Omuta Areas, and Cretaceous Granitic Rocks on North Kyushu Island

Site	(Latitude, Longitude)	Rock	N	Levels	D(°)	I(°)	D^*(°)	I^*(°)	α_{95}(°)	k
				Sasebo-Imari Area						
Nojima Group (Oya Formation)										
5	(33°13.1', 129°34.0')	silt	6	L 360°-620°C	205.8	-39.6	188.0	-44.4	6.0	127.6
6	(33°13.1', 129°34.0')	silt	4	L 260°-440°C	2.5	41.1	345.3	39.6	18.6	25.4
Sasebo Group										
13	(33°26.1', 129°48.8')	tuff	4	L 10-25 mT	353.8	59.8	336.5	54.9	18.1	26.8
Ainoura Group										
12	(33°16.7', 129°50.8')	mud stone	6	L 200°-320°C	27.3	61.5	4.3	62.0	19.8	12.4
Kishima Group										
10	(33°10.5', 129°51.8')	tuff	6	L 300°-600°C	355.5	45.0	0.6	54.0	7.5	151.6
11	(33°10.5', 129°51.8')	tuff	4	L 5-20 mT (2), G 0-12.5 mT (2)	163.4	-48.6	163.2	-58.6	4.9	617.3
	Mean		[6]		4.4	50.4			12.4 [ξ2=3.770]	30.3
	Mean*		[6]				353.0	52.9	9.7 [ξ2=0 437]	48.8
				Kita-Kyushu Area						
Ashiya Group										
2	(33°55.5', 130°44.1')	mud stone	7	L 220°-280°C (2), G 120°-280°C (5)	235.1	-15.8	218.1	-51.6	6.0	125.4
3	(33°56.1', 130°42.8')	mud stone	7	L 280°-320°C (2), G 120°-320°C (5)	247.1	-1.6	225.8	11.3	5.1	173.3
5	(33°56.1', 130°42.3')	silt	8	L 200°-340°C (5), G 120°-280°C (3)	245.3	-65.2	234.0	-52.2	6.2	85.5
8	(33°54.7', 130°40.2')	mud stone	6	L 200°-360°C (4), G 120°-320°C (2)	189.3	-70.1	217.0	-52.0	4.8	209.0
9	(33°54.7', 130°40.2')	mud stone	9	L 10-20 mT (5), G 0-20 mT (4)	46.1	57.8	50.4	36.1	5.8	83.9
16	(33°53.5', 130°42.3')	tuff	5	L 10-20 mT (3), G 2-20 mT (2)	2.1	51.3	33.8	39.5	8.0	106.6
17	(33°52.9', 130°42.3')	silt	5	L 10-25 mT (3), G 0-25 mT (2)	26.5	56.8	34.2	43.2	8.8	87.2
18	(33°53.1', 130°41.2')	tuff	5	L 320°-440°C (2), G 120°-320°C (3)	173.8	-49.4	187.9	-36.1	2.2	1611.9
	Mean		[7]	(except site 3)	22.7	54.2			16.4 [ξ2=4.959]	14.5
	Mean*		[7]	(except site 3)			35.9	45.3	9.9 [ξ2=1.496]	37.8
				Amakusa Area						
Akasaki Group										
1	(32°39.3', 130°31.2')	red silt	4	L 590°-680°C	203.7	-51.8	194.2	-59.4	6.1	226.1
4	(32°30.2', 130°25.4')	red silt	7	L 360°-650°C	220.9	-47.8	210.1	-46.6	3.2	357.7
5	(32°23.3', 130°21.3')	red silt	6	L 360°-650°C	210.5	-52.2	191.5	-26.0	3.4	387.0
6	(32°23.3', 130°21.2')	red silt	5	L 560°-650°C	217.8	-49.9	189.5	-29.2	9.0	73.7
14	(32°33.9', 130°28.6')	red silt	6	L 590°-680°C	196.3	-49.8	225.1	-57.1	2.3	886.4
	Mean		[5]		210.0	-50.7			6.4	145.5
	Mean*		[5]				200.1	-44.5	17.9	19.2
				Omuta Area						
Ginsui Formation										
1	(33°05.1', 130°31.0')	red silt	6	L 580°-660°C	210.0	-44.2	215.3	-68.0	6.0	127.2
				Cretaceous Granitic Rocks						
4	(32°39.3', 130°31.2')		10	L 500°-560°C	83.8	45.2			5.1	91.7
5	(32°30.2', 130°25.4')		7	L 500°-540°C	46.3	71.2			19.1	11.0
6	(32°23.3', 130°21.3')		6	L 25-60 mT	82.1	54.6			10.8	39.8
7	(32°23.3', 130°21.2')		8	L 500°-540°C	38.3	37.7			19.4	9.1
	Mean		[4]		63.4	54.1			24.1	15.5

Site is site number and latitude (N) and longitude (E) of site locality. N, number of specimens [sites]; Levels, demagnetization levels included in least squares line fitting toward the origin of the vector enfpoint diagram (L) and in least squares great circle fitting (G), respectively; D, I, declination and inclination in in situ coordinates, respectively; D^*, I^*, declination and inclination after tilt correction, respectively; α_{95}, radius of 95% confidence circle; k, precision parameter; ξ2, test statistic for the fold test of *McFadden* [1990].

negative polarity, close to the in situ directions of the secondary components from the Akasaki Group in the Amakusa area (Figure 5 and Table 1). There is no stable paleomagnetic data in the Omuta area except site 1, so it was not possible to determine whether or not the ChRM of site 1 is primary. The remanent direction of site 1 is omitted from further consideration.

Cretaceous granitic rocks. NRMs of the granite specimens were generally characterized by broad unblocking temperatures below 580°C and by low MDFs of about 5 mT. NRM intensities of most specimens decayed to less than 10- 20% of initial values by about 400°C and/or 15 mT, and erratic changes in direction occurred at higher demagnetization levels. Below 400°C and/or

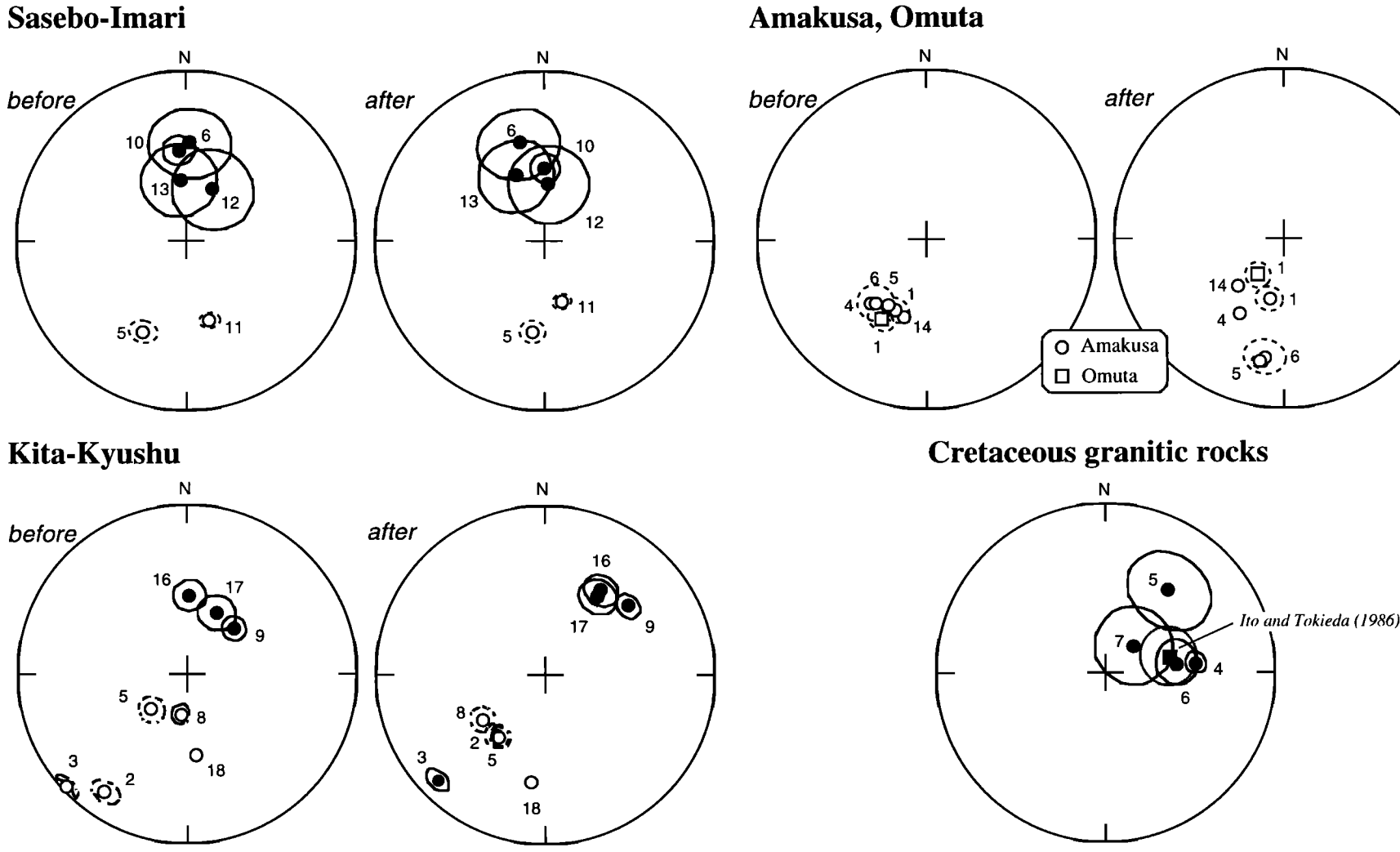


Figure 5. Equal-area projections of site mean directions before and after tilt corrections. Solid (open) symbols are on the lower (upper) hemispheres, respectively. Ovals around the directions indicate 95% confidence limit. Numerals denote site numbers.

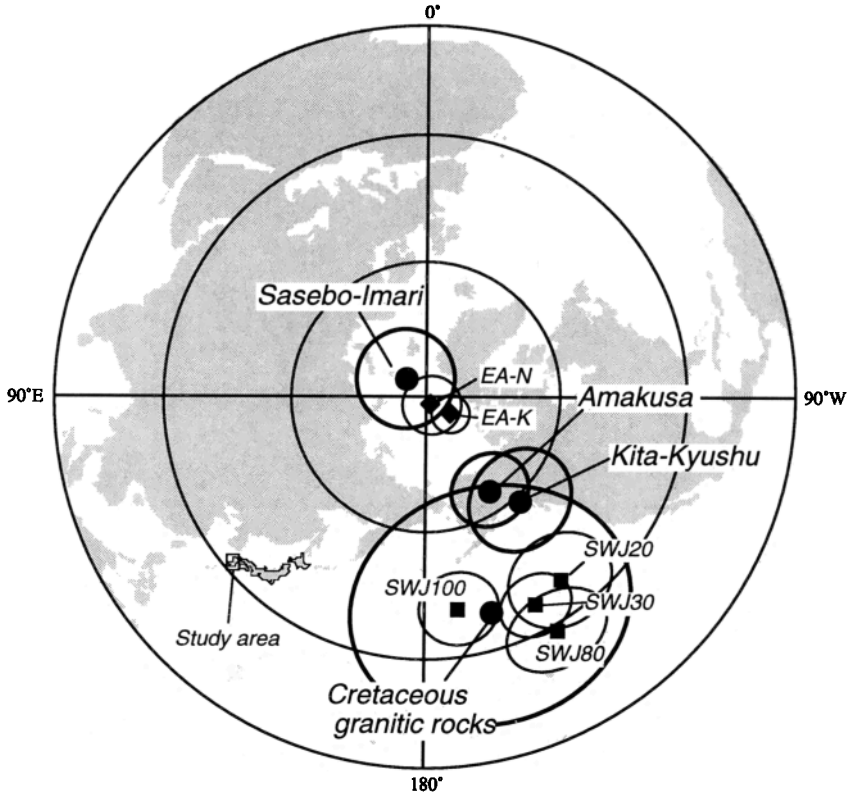


Figure 6. Equal-area projection of paleomagnetic poles for the Sasebo-Imari and Kita-Kyushu areas. Pole positions determined from secondary magnetizations of the Akasaki Group in the Amakusa area and from Cretaceous granitic rocks are also shown. EA-N and EA-K, Neogene and Cretaceous paleomagnetic poles of eastern Asia [*Otofuji et al.*, 1995], respectively; SWJ20, SWJ30, SWJ80, and SWJ100, paleomagnetic poles of 20, 30, 80 and 100 Ma for the main part of Southwest Japan [*Otofuji and Matsuda*, 1987], respectively.

15 mT, these specimens showed unstable magnetic behavior or linear trends of vector endpoints which did not decay toward the origin of the diagrams. Specimens from four of 13 sites, however, yielded ChRMs at demagnetization steps above approximately 500°C and/or 20 mT (Figure 4c). Based on progressive demagnetization results, the ChRMs of these four sites are probably carried by small grains of Ti-poor titanomagnetite, whereas the principal carriers of NRMs for the other sites may be titanomagnetites of multidomain size. Site mean directions of the ChRMs from the four sites show easterly-deflected directions with normal polarity in in situ coordinates (Table 1 and Figure 5). The directions are consistent with those of *Ito and Tokieda* [1986]; their mean direction is D=75.6°, I=58.1°, α_{95}=14.1° and k=15.1. However, these authors do not give the results of demagnetization experiments which they made to obtain the stable magnetic component. A mean direction calculated using the in situ site mean directions in this study is D=67.4°, I=54.4°, α_{95}=24.5° and k=15.1. There is no information that can be used for structural correction of the granitic rocks.

Discussion

Pole Positions and Estimation of Tectonic Displacements

Paleomagnetic pole positions for the Sasebo-Imari and Kita-Kyushu areas are shown in Figure 6, as are those for eastern Asia [*Bretstein*, 1988; *Zhao et al.*, 1990; *Zheng et al.*, 1991; *Otofuji et al.*, 1995] and the main part of SW Japan [*Otofuji and Matsuda*, 1987]. The poles for the Sasebo-Imari and Kita-Kyushu areas were determined from paleomagnetic data covering stratigraphically long periods relative to paleosecular (Figure 3). Thus, the pole positions for the two areas are probably free from the effect of paleosecular variation, although the number of the data used in calculating the pole positions is slightly smaller than that recommended by *Butler* [1991]. These pole positions are inferred to be the result of tectonic movement within the two areas. Paleomagnetic poles from eastern Asia are indispensable in estimating the amount of tectonic displacement that the Japanese Islands were subjected to during formation of the Japan Sea. The number of reliable paleopoles is, however, still few. A mean Neogene paleopole (87.6°N, 146.5°W, α_{95}=6.7) calculated from the poles of *Bretstein* [1988], *Zhao et al.* [1990], and *Zheng et al.* [1991] is indistinguishable from the geographic north pole, and the Cretaceous pole compiled by *Otofuji et al.* [1995] is close to the geographic north pole as well (Figure 6). No reliable Paleogene pole has been obtained from eastern Asia [*Enkin et al.*, 1992]. Therefore, taking the geographic north pole as a reference, apparent tectonic displacements of the Sasebo-Imari and Kita-Kyushu areas relative to eastern Asia are calculated from the paleomagnetic directions of these two areas (Table 2).

The Sasebo-Imari pole overlaps the geographic north pole (Figure 6) and shows no significant discrepancies in either declination or inclination (Table 2). It is therefore suggested that the Sasebo-Imari area has not been displaced significantly relative to eastern Asia since about 20 Ma. The significant discordance between the Sasebo-Imari pole and the reference

Table 2. Paleomagnetic Data From North Kyushu Island and Its Adjacent Areas

Area / Rock Unit	*Location*			*Direction* (°)					*Pole Position*				*Tectonic Parameters*				
	°N	°E	*N*	*Dm*	*Im*	α_{95}(°)	*k*	*ΔD*(°)	°N	°E	A_{95}(°)	*K*	*R*(°)	*ΔR*(°)	*F*(°)	*ΔF*(°)	R.P.
Sasebo-Imari area	33.25	129.82	6	-7.0	52.9	9.7	48.8	16.2	84.1	49.9	11.1	37.2	-7.0	12.7	-0.2	7.6	GNP
													-9.9	14.1	0.0	9.3	EA-N
													-66.1	15.7	-11.4	13.7	SWJ20
Kita-Kyushu area	34.90	130.68	7	35.9	45.3	9 9	37.8	14.1	58.2	-137.0	11.1	30.7	35.9	11.0	9.9	7.7	GNP
													34.9	12.8	9.3	9.3	EA-N
													-28.2	12.8	2.0	10.3	SWJ30
Amakusa area (in situ)	32.52	130.42	5	30.0	50.7	6.4	145.5	10.1	64.7	-145.2	8.4	84.5	30.0	7.8	1.2	4.9	GNP
Creatceous granitic rocks	33.00	130.25	4	63.4	54.1	24 1	15.5	44.1	38.8	197.6	28 9	11.1	55.8	33.8	-3.6	18.7	EA-K
													-7 1	34 5	-14.1	21.0	SWJ80
						Main Kyushu							[Reference]				
Ono volcanic rocks	33.01	131.50	3	2.3	50.3	25 8	23.8	43.0	88.6	-89.0	28 6	19.6	[Torii and Ishıkawa (1986)]*				
Sobosan-Okueyama volcano-plutonic complex	32.70	131.40	4	-2.0	53.9	13.0	50.6	22.4	87.2	101.8	15.8	35.0	[Torii and Ishikawa (1986)]				
						Western Kyushu											
Kıta-Matsuura basalts	33.30	129.80	9	-5.1	46.8	5.0	108.8	7.3	83.4	-9.5	5.7	82 0	[Ozima et al. (1968)]**				
						Tsushıma Strait Area											
Goto Islands																	
Goto Group [G]	32.90	128.90	4	-38.2	40.7	23.0	16.9	31.0	56.4	33.3	24.4	15.1	[Ishikawa and Tagami (1991)]***				
Middle Miocene volcanic rocks [V]	32.90	128.90	7	-2.3	45.4	8.2	55.1	11.7	84.1	-31.8	9.3	43.5	[Ishikawa and Tagami (1991)]				
Tsushima Islands	34.30	129.30	6	-22.4	47.1	11.7	33 9	17.3	65.4	47.0	23.8	8.9	[Ishikawa et al. (1989)]				

N, number of the data used in calculating mean directions; *Dm* and *Im* , declination and inclination of the observed mean directions, respectively, *ΔD*, 95% confidence limit in declination ($\Delta D=\sin^{-1}[\sin(\alpha_{95})/\cos(I)]$); *Pole positions*, latitude (N) and longitude (E) of the north seeking pole positions;. A_{95} and *K*, a radius of 95% confidence circle and precision parameter of the pole, respectively; *Tectonic parameters*, discrepancies in declination (*R*) and inclination (*F*) between the observed and expected directions and their 95% uncertainties (*ΔR*, *ΔF*) are calculated based on the definitions of *Beck* [1980] and *Demarest* [1983]; R.P., reference poles from which expected directions are calculated; GNP, geographic north pole; SWJ20, SWJ30, and SWJ80, 20, 30, and 80 Ma paleomagnetic poles for SW Japan, respectively [*Otofuji and Matsuda*, 1987]; EA-N and EA-K, Neogene and Cretaceous paleomagnetic poles for eastern Asia [*Otofuji et al.*, 1995], respectively.

*Mean direction of Ono volcanic rocks was recalculated from site mean directions of *Torii and Ishikawa* [1986].

**Kita-Matsuura basalts represent a mean recalculated from remanent directions of *Ozima et al.* [1968] with smaller α_{95} than 20°.

***Mean direction of the Goto Group was calculated from *Ishıkawa and Tagami* [1991].

poles for the main part of SW Japan is attributed to middle Miocene CW rotation of SW Japan. On the other hand, the Kita-Kyushu pole is different from the geographic north pole and the 30 Ma pole of the main part of SW Japan (Figure 6). The data indicate a 35.9° ± 11.0° CW rotation of the Kita-Kyushu area relative to eastern Asia and a CCW rotation of 28.2° ± 12.8° relative to the main part of SW Japan since about 30 Ma (Table 2). The discordance between the Kita-Kyushu and geographic north poles also provides a discrepancy of 9.9° ± 7.7° in inclination. This discrepancy could imply northward translation of the Kita- Kyushu area relative to eastern Asia, but it may also be due to inclination shallowing. However, using the Neogene pole of eastern Asia as a reference instead of the geographic north pole, the discrepancy in inclination is not statistically significant (Table 1). In order to establish its significance, more paleomagnetic data from the region around the Kita- Kyushu area are needed as well as a reliable Paleogene pole for eastern Asia.

The pole position for secondary remanences from the Akasaki Group in the Amakusa area is also different from the geographic north pole (Figure 6). The directions of these secondary magnetizations are tightly clustered with a reversed polarity (Figure 5), implying the possibility that the pole position of the Akasaki Group may represent a temporal position of the geomagnetic pole. Following *Cox* [1969], the angular dispersion (*S*) of the corresponding virtual geomagnetic poles (VGPs) is 8.8° with the lower and upper 95% confidence limits (*Sl* and *Su*) of 6.0° and 16.9°, respectively. The mean *S* of data for last 5 Ma in the latitude ranges between 30° and 40° is 15.5° with *Sl* and *Su* values of 14.5° and 16.6°, respectively, while that for the interval 5-45 Ma at latitudes between 30° and 45° is 21.0° with *Sl* and *Su* of 20.4° and 22.9°, respectively [*McFadden and McElhinny*, 1984]. The *S* value for the Amakusa area is smaller than expected. It is thus suggested that paleosecular variation has not been averaged out in calculating the mean pole of the Akasaki Group. The above-mentioned possibility remains. Alternatively, if the discrepancy between the mean pole and the geographic north pole is attributed to tectonic movements, a regional CW rotation of 30.0° ± 7.8° about a vertical axis is indicated (Table 2). A regional tilting of 24° toward the WNW direction about a horizontal axis striking N12°E may also explain the discrepancy. Rotations of small blocks instead of regional movement may also be a possible explanation for the discrepancy [e.g., *MacDonald*, 1980]. However, the geologic structures in the Amakusa area do not support either tilting of the whole Amakusa area or rotations of small blocks: the folds in the Amakusa area show no significant plunge, and although there is some evidence for faults cutting the fold system, the structural trends in the Amakusa area do not change significantly between inferred blocks, except at the Uto Peninsula [*Miki*, 1975]. A regional CW rotation of the whole Amakusa area may be the most reasonable tectonic explanation for the deflected Amakusa pole. The Amakusa pole is different from the reference poles of SW Japan and is situated close to the Kita-Kyushu pole (Figure 6). The Amakusa and Kita-Kyushu areas might have shared a similar rotational motion.

The pole of Cretaceous granitic rocks is different from the Cretaceous pole of eastern Asia and is close to the 80 and 100 Ma poles of SW Japan (Figure 6), which implies a CW rotation of 55.8° ± 33.8° about a vertical axis relative to eastern Asia and no significant tectonic motion relative to the main part of SW Japan since the Cretaceous time (Table 2). However, the possibility can not be excluded that the granite pole position could be the result of geomagnetic phenomena and/or tectonic tilting because the pole position of the granitic rocks is determined from a small number of data with no information for a structural correction. The Kita-Kyushu pole obtained from Paleogene sedimentary rocks overlying the Cretaceous granitic rocks indicates significant CW and CCW rotations since 30 Ma relative to eastern Asia and the main part of SW Japan, respectively. The circle of 95% confidence for the granite pole includes the Kita-Kyushu pole (Figure 6). It is thus suggested that the granite pole position may be attributed at least partially to the rotational motion indicated by the Kita-Kyushu pole.

Rotational Motions in North Kyushu Island Related to CW Rotation of SW Japan

Paleomagnetic declination data from north Kyushu Island and its neighboring regions are shown in Figure 7. The deflection from north in declination represents the sense and amount of rotation. Two distinct tectonic blocks are inferred to exist in north Kyushu Island based on the discordant declinations: the unrotated western part of north Kyushu Island (Western Kyushu) and the rotated main part of north Kyushu Island (Main Kyushu).

In Western Kyushu, non deflected paleomagnetic directions are observed in Tertiary sedimentary rocks and the overlaying late Miocene Kita-Matsuura basalts [*Creer and Ispir*, 1970; *Nomura*, 1967; *Ozima et al.*, 1968]. No significant rotation relative to eastern Asia has taken place since about 20 Ma in Western Kyushu. On the other hand, Main Kyushu features both CW-deflected and non deflected directions of different ages. Paleomagnetic data from the Kita-Kyushu area indicate a CW rotation of 36° relative to eastern Asia since about 30 Ma. The CW rotation probably affected all of Main Kyushu because pre-Miocene units show approximately consistent structural trends. Non deflected paleomagnetic directions come from the middle Miocene Ono volcanic rocks and Sobosan-Okueyama volcano-plutonic complex [*Torii and Ishikawa*, 1986]. These igneous rocks intrude into and/or overlie pre-Miocene units, and have K-Ar ages of 13-14 Ma [*Shibata and Ono*, 1974; *Shibata*, 1978; *Tatsumi et al.*, 1980]. It is thus suggested that CW rotation of Main Kyushu occurred between 30 and 14 Ma. CW deflections of remanent directions from the Cretaceous granitic rocks and the Akasaki Group in the Amakusa area are consistent with this CW rotation.

SW Japan has been regarded as a coherent rigid block that was subjected to CW rotation of approximately 50° at about 15 Ma [e.g., *Otofuji and Matsuda*, 1987]. However, the derived rotational motions of Main Kyushu and Western Kyushu imply that neither region is a part of the coherent SW Japan block. The zonal structure of pre-Miocene units in the main part of SW Japan, which has been considered an indicator of the extent of the CW-rotated SW Japan block [e.g., *Otofuji and Matsuda*, 1987], can be continuously traced into Main Kyushu. The CW rotation of Main Kyushu is probably associated with that of SW Japan at about 15 Ma, although Main Kyushu did not rotate as a part of the coherent SW Japan block. The paleomagnetic data from Main Kyushu indicate that this region underwent a significant CCW rotation (28° ± 13°) relative to the main part of SW Japan between 30 and 14 Ma. It can be inferred that this CCW rotation occurred during the CW rotation of the SW Japan block. It is therefore probable that the SW Japan block which experienced middle Miocene CW rotation is truncated west of Main Kyushu, and that Main Kyushu, representing the westernmost part of the SW Japan block, did not rotate as much as the main part of the SW Japan block during its CW rotation of the whole block. *Murata* [1987a, b] and *Kano et al.* [1990] show that there is a difference in the structural trend of pre-Miocene units between

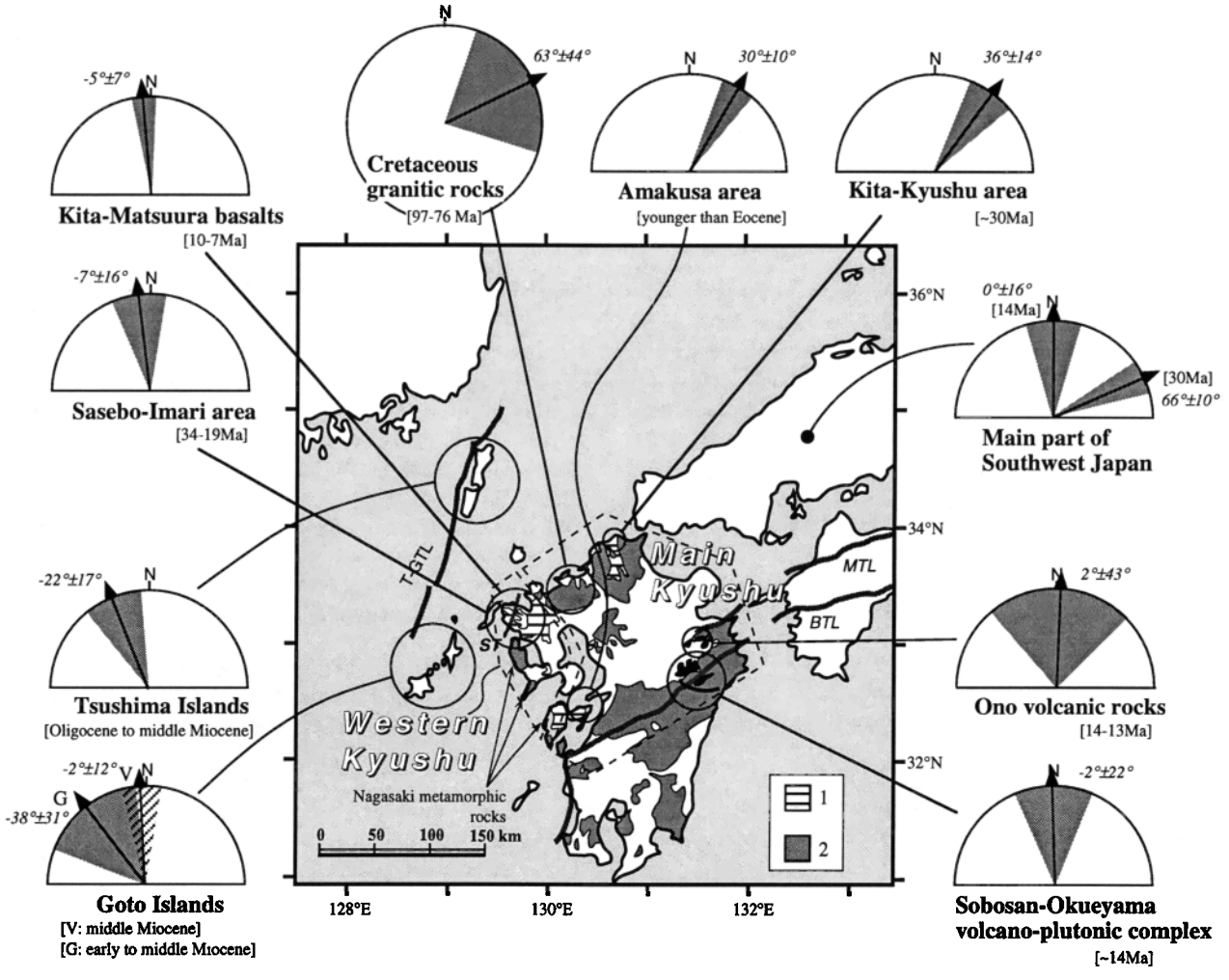


Figure 7. Paleomagnetic declinations on north Kyushu Island and its adjacent areas with 95% confidence limits (shaded) listed in Table 2. Declinations of 30 and 14 Ma in the main part of SW Japan are after *Otofuji and Matsuda* [1987] and *Otofuji et al.* [1991], respectively. 1, Paleogene to Miocene sedimentary rocks on north Kyushu Island; 2, pre-Miocene geotectonic units in the Kyushu Island [after *Geological Survey of Japan,* 1987]; ST, Sasagawa thrust; MTL, Median tectonic line; BTL, Butsuzo tectonic line; TGTL, Tsushima-Goto tectonic line.

Main Kyushu (N55° ± 5°E) and the main part of SW Japan (N75° ± 5°E). The discrepancy in the structural trends provides further evidence for differential rotation of Main Kyushu. However, no large scale fault zone is observed between the main part of SW Japan and Main Kyushu, and the SW Japan block probably behaved as a non rigid body during its CW rotation.

In Western Kyushu, geological structures of the Nagasaki metamorphic rocks show N-S to NE-SW trends, which are different from the structural trend of metamorphic belts in SW Japan [*Hattori and Isomi,* 1976; *Faure et al.*, 1988; *Nishiyama,* 1989; *Miyazaki and Nishiyama,* 1989]. The Nagasaki metamorphic rocks are correlated with the Sanbagawa metamorphic rocks [*Faure et al.*, 1988] or the Sangun metamorphic rocks in SW Japan [*Hattori and Shibata,* 1982]. Regardless of whether the Nagasaki metamorphic belt is correlated with the Sanbagawa or Sangun belts, the fact that the trends of the Nagasaki belt differs from the common trend of the other belts in SW Japan indicates that the Nagasaki belt has behaved as an independent block relative to SW Japan [*Kizaki,* 1979, 1986; *Faure et al.* 1988]. The paleomagnetic data indicate that Western Kyushu did not undergo the CW rotation of SW Japan. Also, it is confirmed that Western Kyushu behaved as a different tectonic block from SW Japan in middle Miocene time. The significant difference in the structural trend between pre-Miocene units in Western Kyushu and SW Japan may imply the existence of a fault zone between these two regions [*Hattori and Shibata,* 1982]. The Hatashima Fault [*Kobayashi et al.,* 1956] separates Tertiary sedimentary sequences in the Sasebo-Imari area from Cretaceous granites, which yield CW-deflected remanent directions. *Miki* [1975] and *Hattori and Shibata* [1982] suggested that Tertiary sedimentary basins in Western Kyushu were formed as grabens. The faults forming these basins might have accommodated differential rotation between Western Kyushu and SW Japan during CW rotation of SW Japan.

Ishikawa and Tagami [1991] suggested that compressive deformation took place in the Tsushima Strait area in early to middle Miocene time, approximately coeval with CW rotation of SW Japan. They assumed that the CW-rotated SW Japan block included the whole of north Kyushu Island and proposed a model that the compressive deformation was caused by convergence between the western margin of the SW Japan block and the Korean Peninsula as a result of CW rotation of the block about a

pivot on its western part. The present work suggests that Western Kyushu has been subjected to no significant tectonic displacement since 20 Ma, although Main Kyushu was rotated CW as the westernmost part of the SW Japan block. The NE-SW trending Sasagawa thrust in the Sasebo-Imari area (Figure 7) was active in early to middle Miocene time [*Nagahama*, 1965]. The sense and timing of the thrust movement may imply a northwestward motion in Western Kyushu during the compressive deformation in the Tsushima Strait area, although our paleomagnetic data indicate no tectonic displacement of Western Kyushu. The amount of the northwestward motion of Western Kyushu, if such motion had occurred, should have been smaller than that which can be resolved by paleomagnetic techniques. The model of *Ishikawa and Tagami* [1991] may be modified as follows: the CW rotation of the SW Japan block about a pivot on its western part resulted in convergence between the western tip of the SW Japan block and the Korean Peninsula and subsequent northwestward motion of Western Kyushu.

Conclusions

Samples from Tertiary sedimentary rocks and Cretaceous granitic rocks in north Kyushu Island were studied paleomagnetically in order to clarify the tectonic motions of north Kyushu Island and relate them to the documented middle Miocene CW rotation of SW Japan. A paleomagnetic direction determined from Oligocene sedimentary rocks in the northern part of north Kyushu Island indicates a significant CW rotation of 36° relative to eastern Asia since 30 Ma, while a paleomagnetic direction from Oligocene to early Miocene sedimentary rocks in the western part of north Kyushu Island shows no significant displacement relative to eastern Asia since 20 Ma. Eocene sedimentary rocks in the southwestern part of north Kyushu Island yield stable secondary remanences with CW- deflected in situ directions. In situ directions of stable remanences from Cretaceous granitic rocks also show CW- deflected declinations. These paleomagnetic data, and previously reported data, suggest that the main part of north Kyushu Island (Main Kyushu) was subjected to a CW rotation of 36° ± 11° between 30 and 14 Ma relative to eastern Asia, whereas no significant tectonic displacements have taken place in the western part (Western Kyushu) since 20 Ma. Taking into account regional geological structures, the paleomagnetic results indicate that the SW Japan block, which experienced middle Miocene CW rotation, is extended as far as the west of Main Kyushu. The paleomagnetic data also indicate that Main Kyushu was subjected to a significant CCW rotation of 28° ± 13° relative to the main part of SW Japan between 30 and 14 Ma. The differential rotation implies that the westernmost part of the SW Japan block underwent a smaller amount of CW rotation during the rotation of the whole of SW Japan.

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